

Lecture 5: Second derivatives and convex function

MA3236 Non-Linear Programming

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Quick review

Let $C \subseteq \mathbf{R}^n$ be convex, and let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ be differentiable.

- If f is convex on C , every local minimizer is global.
If f is strictly convex on C , there is at most one global minimizer.

Quick review

Let $C \subseteq \mathbf{R}^n$ be convex, and let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ be differentiable.

- If f is convex on C , every local minimizer is global.
If f is strictly convex on C , there is at most one global minimizer.
- f is convex on $C \Leftrightarrow$ its graph lies above every tangent plane over C .

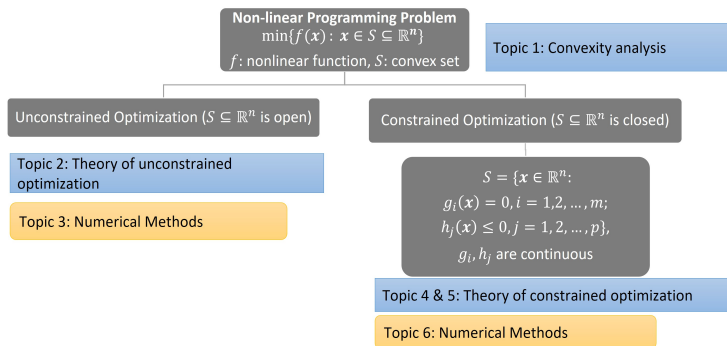
$$f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}), \quad \forall \mathbf{x}, \mathbf{y} \in C.$$

- Necessary and sufficient condition for minimizing convex f over C :

$$\nabla f(\mathbf{x}^*)^T (\mathbf{x} - \mathbf{x}^*) \geq 0, \quad \forall \mathbf{x} \in C.$$

If C is open (in particular, if $C = \mathbf{R}^n$), then $\nabla f(\mathbf{x}^*) = 0$.

Topics



In lectures 3–5, we will study convex analysis (Topic 1).

- Convex sets and convex functions.
- First-order characterizations of convex functions.
- Second-order characterizations of convex functions.

Hessian matrix

So far, we have learnt to prove a function is convex/strictly convex by

- definition
- first-order tangent plane characterization

These characterizations are not always convenient in concrete examples.

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Definition 5.1 (Hessian matrix)

Let $f : S \rightarrow \mathbf{R}$, where $S \subseteq \mathbf{R}^n$ is a nonempty set. Suppose $\mathbf{x}$ is an interior point of S and the second partial derivatives of f are continuous in a neighborhood of $\mathbf{x}$. The **Hessian** of f at $\mathbf{x}$ is the $n \times n$ matrix

$$H_f(\mathbf{x}) := \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n}(\mathbf{x}) \end{pmatrix}.$$

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Example 5.2

(a) Define $f : \mathbf{R} \rightarrow \mathbf{R}$ by $f(x) := x^3 + 2x + e^x$. Then

$$f'(x) = 3x^2 + 2 + e^x, \quad H_f(x) = f''(x) = 6x + e^x.$$

(b) Define $f : \mathbf{R}^2 \rightarrow \mathbf{R}$ by $f(\mathbf{x}) := x_1^3 + 2x_1x_2 + x_2^2$. Then

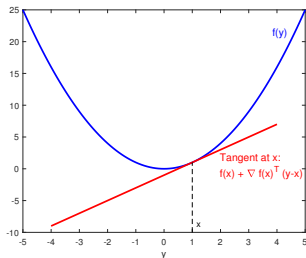
$$\nabla f(\mathbf{x}) = \begin{pmatrix} 3x_1^2 + 2x_2 \\ 2x_1 + 2x_2 \end{pmatrix}, \quad H_f(\mathbf{x}) = \begin{pmatrix} 6x_1 & 2 \\ 2 & 2 \end{pmatrix}.$$

Hessian and convexity: one-dimensional case

Let $f : \mathbf{R} \rightarrow \mathbf{R}$ have a continuous second derivative and $t, \bar{t} \in \mathbf{R}$.

$$f(\bar{t}) = f(t) + f'(t)(\bar{t} - t) + \frac{1}{2}f''(t)(\bar{t} - t)^2 + o((\bar{t} - t)^2).$$

$$f \text{ convex} \Leftrightarrow f(\bar{t}) \geq f(t) + f'(t)(\bar{t} - t)$$



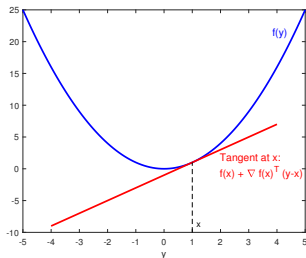
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$$\Leftrightarrow f''(t)(\bar{t} - t)^2 + o((\bar{t} - t)^2) \geq 0 \Leftrightarrow f''(t) \geq 0 \quad \forall t \in \mathbf{R}$$



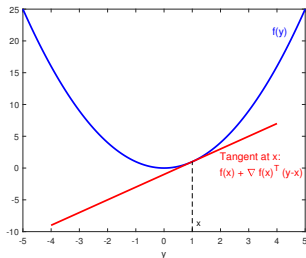
$f''(t)$: curvature at t

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$$f''(t) > 0 \quad \forall t \in \mathbf{R} \quad \Rightarrow \quad f \text{ strictly convex}$$



$f''(t)$: curvature at t

Hessian and convexity: multi-dimensional case

Let $S \subseteq \mathbf{R}^n$ be open, $f : S \rightarrow \mathbf{R}$ have continuous second partial derivatives, and $\mathbf{x} \in S$. For $\mathbf{d} := (d_1, \dots, d_n)^T \in \mathbf{R}^n \setminus \{\mathbf{0}\}$, define

$$\phi(t) := f(\mathbf{x} + t\mathbf{d})$$

for t sufficiently close to 0. The chain rule gives

$$\phi'(0) = \nabla f(\mathbf{x})^T \mathbf{d}, \quad \phi''(0) = \mathbf{d}^T H_f(\mathbf{x}) \mathbf{d}.$$

Hessian and convexity: multi-dimensional case

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$\mathbf{d}^T H_f(\mathbf{x}) \mathbf{d}$ is the second directional derivative (curvature) at $\mathbf{x}$ along $\mathbf{d}$:

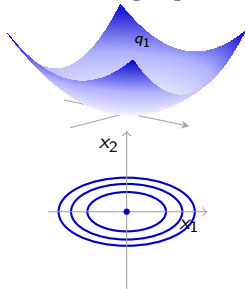
- $\mathbf{d}^T H_f(\mathbf{x}) \mathbf{d} > 0$: upward curvature along $\mathbf{d}$;
- $\mathbf{d}^T H_f(\mathbf{x}) \mathbf{d} < 0$: downward curvature along $\mathbf{d}$;
- $\mathbf{d}^T H_f(\mathbf{x}) \mathbf{d} = 0$: zero second-order term along $\mathbf{d}$.

Directional curvature shapes quadratic level curves

Strictly convex

$$q_1(\mathbf{x}) := \frac{1}{2}(x_1^2 + 4x_2^2)$$

levels $0, \frac{1}{2}, 1, \frac{3}{2}$

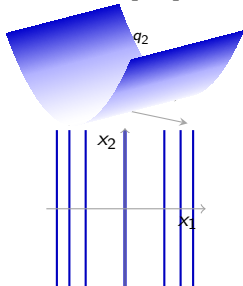


$$\mathbf{d}^T H_{q_1} \mathbf{d} = d_1^2 + 4d_2^2 \\ \geq \|\mathbf{d}\|^2$$

Convex, not strictly convex

$$q_2(\mathbf{x}) := \frac{1}{2}x_1^2$$

levels $0, \frac{1}{2}, 1, \frac{3}{2}$

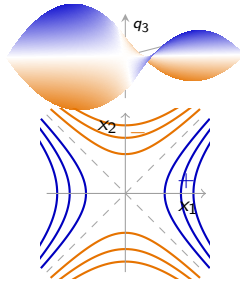


$$\mathbf{d}^T H_{q_2} \mathbf{d} = d_1^2 \geq 0 \\ = 0 \text{ for } \mathbf{d} = (0, 1)^T$$

Nonconvex

$$q_3(\mathbf{x}) := \frac{1}{2}(x_1^2 - x_2^2)$$

levels $0, \pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}$



$$\mathbf{d}^T H_{q_3} \mathbf{d} = d_1^2 - d_2^2 \\ \text{takes both signs}$$

Second derivative test for convexity

Theorem 5.3

Let $C \subseteq \mathbf{R}^n$ be open and convex, and let $f : C \rightarrow \mathbf{R}$ have continuous second partial derivatives.

Hessian $H_f(\mathbf{x})$	Relation	function f
$\mathbf{d}^T H_f(\mathbf{x}) \mathbf{d} \geq 0 \quad \forall \mathbf{d} \in \mathbf{R}^n, \mathbf{x} \in C$	$\Leftrightarrow$	convex

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$\mathbf{d}^T H_f(\mathbf{x}) \mathbf{d} > 0 \quad \forall \mathbf{0} \neq \mathbf{d} \in \mathbf{R}^n, \mathbf{x} \in C$	$\Rightarrow$	strictly convex

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$\mathbf{d}^T H_f(\mathbf{x}) \mathbf{d} \leq 0 \quad \forall \mathbf{d} \in \mathbf{R}^n, \mathbf{x} \in C$	$\Leftrightarrow$	concave
$\mathbf{d}^T H_f(\mathbf{x}) \mathbf{d} < 0 \quad \forall \mathbf{0} \neq \mathbf{d} \in \mathbf{R}^n, \mathbf{x} \in C$	$\Rightarrow$	strictly concave

WARNING. Checking the Hessian at only one point is not sufficient. For example, define $g : \mathbf{R} \rightarrow \mathbf{R}$ by

$$g(x) := x^4 - x^2.$$

then $g''(1) = 10 > 0$, $g''(0) = -2 < 0$, so g is not convex on $\mathbf{R}$.

Positive (semi)definite matrices

Definition 5.4

Let $\mathbf{A}$ be a real symmetric $n \times n$ matrix.

- (a) $\mathbf{A}$ is said to be **positive semidefinite** if $\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$, $\forall \mathbf{x} \in \mathbf{R}^n$.
- (b) $\mathbf{A}$ is said to be **positive definite** if $\mathbf{x}^T \mathbf{A} \mathbf{x} > 0$, $\forall \mathbf{x} \neq \mathbf{0}$.
- (c) $\mathbf{A}$ is said to be **negative semidefinite** if $\mathbf{x}^T \mathbf{A} \mathbf{x} \leq 0$, $\forall \mathbf{x} \in \mathbf{R}^n$; i.e. $-\mathbf{A}$ is positive semidefinite.
- (d) $\mathbf{A}$ is said to be **negative definite** if $\mathbf{x}^T \mathbf{A} \mathbf{x} < 0$, $\forall \mathbf{x} \neq \mathbf{0}$; i.e. $-\mathbf{A}$ is positive definite.
- (e) $\mathbf{A}$ is said to be **indefinite** if $\mathbf{A}$ is neither positive nor negative semidefinite.

Example 5.5

Determine the definiteness of the following matrices.

$$(a) \mathbf{A}_1 := \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$$

$$\begin{aligned} \mathbf{x}^T \mathbf{A}_1 \mathbf{x} &= [x_1, x_2] \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= 2x_1^2 - 2x_1x_2 + 3x_2^2 = 2\left(x_1 - \frac{1}{2}x_2\right)^2 + \frac{5}{2}x_2^2 > 0 \quad \text{for } \mathbf{x} \neq \mathbf{0}. \end{aligned}$$

Thus, $\mathbf{A}_1$ is positive definite.

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Thus, $\mathbf{A}_1$ is positive definite.

$$(b) \mathbf{A}_2 := \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\mathbf{x}^T \mathbf{A}_2 \mathbf{x} = [x_1, x_2] \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 + 4x_1x_2 + x_2^2 = (x_1 + 2x_2)^2 - 3x_2^2.$$

Thus, $\mathbf{A}_2$ is indefinite. (*Why?*)

Revisit: second derivative test for convexity

Theorem 5.3

Let $C \subseteq \mathbf{R}^n$ be open and convex, and let $f : C \rightarrow \mathbf{R}$ have continuous second partial derivatives.

Hessian $H_f(\mathbf{x})$	Relation	function f
positive semidefinite on C	$\Leftrightarrow$	convex
positive definite on C	$\Rightarrow$	strictly convex
negative semidefinite on C	$\Leftrightarrow$	concave
negative definite on C	$\Rightarrow$	strictly concave
indefinite at some $\hat{\mathbf{x}} \in C$	$\Rightarrow$	neither convex nor concave

But wait! We need to check the inequality for any $\mathbf{d} \in \mathbf{R}^n$ and $\mathbf{x} \in C$?

Eigenvalue test for checking definiteness of a matrix

Recall that if $\mathbf{q}_1, \dots, \mathbf{q}_n$ are orthonormal eigenvectors with eigenvalues $\lambda_1, \dots, \lambda_n$, then

$$\mathbf{A} = \sum_{k=1}^n \lambda_k \mathbf{q}_k \mathbf{q}_k^T = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T,$$

where $\mathbf{Q} := (\mathbf{q}_1, \dots, \mathbf{q}_n)$ and $\mathbf{\Lambda} := \text{diag}(\lambda_1, \dots, \lambda_n)$.

Theorem 5.6 (Eigenvalue Test)

Let $\mathbf{A}$ be a real **symmetric** $n \times n$ matrix.

- (a) $\mathbf{A}$ is **positive semidefinite** if and only if every eigenvalue of $\mathbf{A}$ is nonnegative.
- (b) $\mathbf{A}$ is **positive definite** if and only if every eigenvalue of $\mathbf{A}$ is positive.
- (c) $\mathbf{A}$ is **negative semidefinite** if and only if every eigenvalue of $\mathbf{A}$ is nonpositive.
- (d) $\mathbf{A}$ is **negative definite** if and only if every eigenvalue of $\mathbf{A}$ is negative.
- (e) $\mathbf{A}$ is **indefinite** if and only if there is a positive eigenvalue of $\mathbf{A}$ and a negative eigenvalue of $\mathbf{A}$.

Eigenvalue Test:

(a) **A** is **positive semidefinite** if and only if every eigenvalue of **A** is nonnegative.

Proof.

“ $\Rightarrow$ ”

Suppose **A** is positive semidefinite. Let λ be an eigenvalue of **A** with nonzero eigenvector **q**. Then $\mathbf{Aq} = \lambda \mathbf{q}$. Thus $0 \leq \mathbf{q}^T \mathbf{Aq} = \lambda \mathbf{q}^T \mathbf{q}$. Since $\mathbf{q}^T \mathbf{q} = \|\mathbf{q}\|^2 > 0$, we have $\lambda \geq 0$.

Eigenvalue Test:

(a) **A** is **positive semidefinite** if and only if every eigenvalue of **A** is nonnegative.

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Suppose **A** is positive semidefinite. Let λ be an eigenvalue of **A** with nonzero eigenvector **q**. Then $\mathbf{A}\mathbf{q} = \lambda\mathbf{q}$. Thus $0 \leq \mathbf{q}^T \mathbf{A}\mathbf{q} = \lambda \mathbf{q}^T \mathbf{q}$. Since $\mathbf{q}^T \mathbf{q} = \|\mathbf{q}\|^2 > 0$, we have $\lambda \geq 0$.

“ $\Leftarrow$ ”

Conversely, suppose all the eigenvalues of **A** are nonnegative. Consider the eigenvalue decomposition $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$, where $\mathbf{Q} \in \mathbf{R}^{n \times n}$ is orthogonal and $\mathbf{\Lambda} \in \mathbf{R}^{n \times n}$ is diagonal with the eigenvalues of **A** on its diagonal. For any $\mathbf{x} \in \mathbf{R}^n$, define $\mathbf{y} := \mathbf{Q}^T \mathbf{x}$. Then $\mathbf{x}^T \mathbf{A}\mathbf{x} = \mathbf{y}^T \mathbf{\Lambda} \mathbf{y} = \sum_{k=1}^n \lambda_k y_k^2 \geq 0$. □

Example 5.7

(a) $\mathbf{A}_1 := \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$. Find the eigenvalues of $\mathbf{A}_1$:

$$\det(\mathbf{A}_1 - \lambda \mathbf{I}) = \begin{vmatrix} 4 - \lambda & 2 \\ 2 & 1 - \lambda \end{vmatrix} = \lambda^2 - 5\lambda = 0$$

$$\implies \lambda = 0, 5.$$

Thus, $\mathbf{A}_1$ is positive semidefinite.

$$(b) \mathbf{A}_2 := \begin{bmatrix} 4 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -2 \end{bmatrix}.$$

The eigenvalues are approximately 1, -2.3166 , and 4.3166 .

Thus, $\mathbf{A}_2$ is indefinite.

Leading principal minors

For small dimensional matrices, another simpler test for positive (negative) **definite** matrix is via leading principal minors.

Definition 5.8 (Leading principal minors)

Let $\mathbf{A} := [a_{ij}]_{i,j=1}^n$ be a real matrix.

The k th leading principal minor Δ_k of $\mathbf{A}$ is defined by

$$\Delta_k := \det \begin{bmatrix} a_{11} & \cdots & a_{1k} \\ \cdot & \cdots & \cdot \\ \cdot & \cdots & \cdot \\ a_{k1} & \cdots & a_{kk} \end{bmatrix}$$

Sylvester criterion

Theorem 5.9

Suppose $\mathbf{A} \in \mathbf{R}^{n \times n}$ is a symmetric matrix.

- (a) $\mathbf{A}$ is **positive definite** if and only if $\Delta_k > 0$ for all $k = 1, 2, \dots, n$.
- (b) $\mathbf{A}$ is **negative definite** if and only if $(-1)^k \Delta_k > 0$ for all $k = 1, 2, \dots, n$
(i.e. the leading principal minors alternate in signs with $\Delta_1 < 0$).

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(i.e. the leading principal minors alternate in signs with $\Delta_1 < 0$).

WARNING. Nonnegative leading principal minors alone do not imply positive semidefiniteness.

For $\mathbf{A} := \text{diag}(0, -1)$, the two leading principal minors are zero, but $[0, 1]\mathbf{A}[0, 1]^T = -1$.

Leading versus all principal minors

Let $\mathbf{A} \in \mathbf{R}^{n \times n}$. For a nonempty index set $I \subseteq \{1, \dots, n\}$, let $\mathbf{A}[I]$ denote the principal submatrix whose row and column indices belong to I .

- $\det \mathbf{A}[I]$ is a **principal minor**.
- The **leading principal minors** are

$$\Delta_k := \det \mathbf{A}[\{1, \dots, k\}], \quad k = 1, \dots, n.$$

For a real symmetric matrix $\mathbf{A}$,

$\mathbf{A}$ is positive semidefinite $\iff \det \mathbf{A}[I] \geq 0$ for every nonempty I .

Example 5.10

(a) For $\mathbf{A}_1 := \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$, $\Delta_1 = 2$, $\Delta_2 = 3$, and $\Delta_3 = 4$.

Since $\Delta_k > 0 \forall k = 1, 2, 3$, $\mathbf{A}_1$ is positive definite.

Example 5.10

(a) For $\mathbf{A}_1 := \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$, $\Delta_1 = 2$, $\Delta_2 = 3$, and $\Delta_3 = 4$.

Since $\Delta_k > 0 \forall k = 1, 2, 3$, $\mathbf{A}_1$ is positive definite.

(b) For $\mathbf{A}_2 := \begin{bmatrix} -1 & 1 & 1 \\ 1 & -4 & 0 \\ 1 & 0 & -2 \end{bmatrix}$, $\Delta_1 = -1$, $\Delta_2 = 3$, and $\Delta_3 = -2$.

Since $(-1)^k \Delta_k > 0 \forall k = 1, 2, 3$, $\mathbf{A}_2$ is negative definite.

Example 5.11

Define $f : \mathbf{R}^3 \rightarrow \mathbf{R}$ by

$$f(x_1, x_2, x_3) := x_1^2 + x_2^2 + x_3^2 - x_1x_2 + x_2x_3 - x_1x_3.$$

Show that f is strictly convex.

Solution.

Since f is twice differentiable, we use the Hessian matrix to check convexity.

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 2x_1 - x_2 - x_3 \\ 2x_2 - x_1 + x_3 \\ 2x_3 + x_2 - x_1 \end{bmatrix} \quad \text{and} \quad H_f(\mathbf{x}) = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$$

which is positive definite (check: $\Delta_1 = 2$, $\Delta_2 = 3$, $\Delta_3 = 4$). Thus, f is strictly convex on $\mathbf{R}^3$.

Revisit: second derivative test for convexity

Theorem 5.3

Let $C \subseteq \mathbf{R}^n$ be open and convex, and let $f : C \rightarrow \mathbf{R}$ have continuous second partial derivatives.

	Hessian $H_f(\mathbf{x})$	Relation	function f
(a)	positive semidefinite on C	$\Leftrightarrow$	convex
(b)	positive definite on C	$\Rightarrow$	strictly convex
(c)	negative semidefinite on C	$\Leftrightarrow$	concave
(d)	negative definite on C	$\Rightarrow$	strictly concave
(e)	indefinite at some $\hat{\mathbf{x}} \in C$	$\Rightarrow$	neither convex nor concave

(b), (d), and (e) are sufficient but not necessary! Counterexamples?

Counterexamples of necessity

(b,d) Define $f(x) := x^4$. The function is strictly convex on $\mathbf{R}$, but

$$f''(x) = 12x^2, \quad f''(0) = 0,$$

so its Hessian is not positive definite at every point. Likewise, $-f$ is strictly concave although $(-f)''(0) = 0$. Thus, the hypotheses in parts (b) and (d) are not necessary.

(e) Define $f(x) := x^3$. The function is neither convex nor concave on $\mathbf{R}$, but its scalar Hessian $f''(x) = 6x$ is never indefinite at any fixed x .

A special case: quadratic functions

Let $\mathbf{A} \in \mathbf{R}^{n \times n}$ be symmetric, let $\mathbf{b} \in \mathbf{R}^n$, and let $c \in \mathbf{R}$. Define $q : \mathbf{R}^n \rightarrow \mathbf{R}$ by

$$q(\mathbf{x}) := \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c.$$

For $\mathbf{x}, \mathbf{y} \in \mathbf{R}^n$ with $\mathbf{x} \neq \mathbf{y}$ and $\lambda \in (0, 1)$, define $\mathbf{d} := \mathbf{x} - \mathbf{y}$. Direct expansion gives

$$\lambda q(\mathbf{x}) + (1 - \lambda)q(\mathbf{y}) - q(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) = \frac{\lambda(1 - \lambda)}{2} \mathbf{d}^T \mathbf{A} \mathbf{d}.$$

Since $\lambda(1 - \lambda) > 0$ and every nonzero $\mathbf{d}$ can occur as $\mathbf{x} - \mathbf{y}$,

$$\begin{aligned} q \text{ is strictly convex} &\iff \mathbf{d}^T \mathbf{A} \mathbf{d} > 0 \text{ for every } \mathbf{d} \neq \mathbf{0} \\ &\iff \mathbf{A} \text{ is positive definite.} \end{aligned}$$

Example 5.12

Define $f : \mathbf{R}^2 \rightarrow \mathbf{R}$ by $f(\mathbf{x}) := 4x_1^2 + 4x_1x_2 + x_2^2 = (2x_1 + x_2)^2$.

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 8x_1 + 4x_2 \\ 4x_1 + 2x_2 \end{bmatrix}, \quad H_f(\mathbf{x}) = \begin{bmatrix} 8 & 4 \\ 4 & 2 \end{bmatrix}.$$

The eigenvalues of $H_f(\mathbf{x})$ are $\lambda = 0$ and $\lambda = 10$. Thus $H_f(\mathbf{x})$ is positive semidefinite for all $\mathbf{x} \in \mathbf{R}^2$. Hence f is convex on $\mathbf{R}^2$.

Example 5.12

Define $f : \mathbf{R}^2 \rightarrow \mathbf{R}$ by $f(\mathbf{x}) := 4x_1^2 + 4x_1x_2 + x_2^2 = (2x_1 + x_2)^2$.

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The eigenvalues of $H_f(\mathbf{x})$ are $\lambda = 0$ and $\lambda = 10$. Thus $H_f(\mathbf{x})$ is positive semidefinite for all $\mathbf{x} \in \mathbf{R}^2$. Hence f is convex on $\mathbf{R}^2$. Moreover, f is identically zero on the line $\{(t, -2t) : t \in \mathbf{R}\}$. For $t \neq 0$, the strict convex inequality fails for $\mathbf{0}$ and $(t, -2t)^T$, so f is not strictly convex.

Example 5.13

Define

$$S := \{\mathbf{x} \in \mathbf{R}^2 : x_1 > 1/3, x_1^3 + 2x_1x_2 + x_2^2 \leq 10\}.$$

Show that S is convex.

Solution. Define

$$C := \{\mathbf{x} \in \mathbf{R}^2 : x_1 > 1/3\}, \quad f(\mathbf{x}) := x_1^3 + 2x_1x_2 + x_2^2.$$

The set C is convex.

Note that the set $S = \{\mathbf{x} \in C : f(\mathbf{x}) \leq 10\}$.

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The set C is convex.

Note that the set $S = \{\mathbf{x} \in C : f(\mathbf{x}) \leq 10\}$.

The Hessian matrix is

$$H_f(\mathbf{x}) = \begin{bmatrix} 6x_1 & 2 \\ 2 & 2 \end{bmatrix}.$$

For $x_1 > 1/3$, $\Delta_1 = 6x_1 > 0$ and $\Delta_2 = 12x_1 - 4 > 0$.

Thus $H_f(\mathbf{x})$ is positive definite on C . Hence f is strictly convex on C , and its sublevel set S is convex.

From convex to generic cases

Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ have continuous second partial derivatives in a neighborhood of $\mathbf{x}$. For $\mathbf{d} \in \mathbf{R}^n$, define the local quadratic model at $\mathbf{x}$ by

$$m_{\mathbf{x}}(\mathbf{d}) := f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T H_f(\mathbf{x}) \mathbf{d}.$$

Define the remainder by

$$r_{\mathbf{x}}(\mathbf{d}) := f(\mathbf{x} + \mathbf{d}) - m_{\mathbf{x}}(\mathbf{d}), \quad \frac{r_{\mathbf{x}}(\mathbf{d})}{\|\mathbf{d}\|^2} \rightarrow 0 \quad \text{as } \mathbf{d} \rightarrow \mathbf{0}.$$

From convex to generic cases

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If $\nabla f(\mathbf{x}) = \mathbf{0}$, then

$$f(\mathbf{x} + \mathbf{d}) - f(\mathbf{x}) = \frac{1}{2} \mathbf{d}^T H_f(\mathbf{x}) \mathbf{d} + r_{\mathbf{x}}(\mathbf{d}).$$

How does the definiteness of $H_f(\mathbf{x})$ help determine local optimality?

Taylor's Theorem

Let $\mathbf{x}, \mathbf{y} \in \mathbf{R}^n$. Define the line segment joining them by

$$[\mathbf{x}, \mathbf{y}] := \{\mathbf{x} + t(\mathbf{y} - \mathbf{x}) \mid t \in [0, 1]\}.$$

Theorem 5.14 (Taylor's Theorem)

Let $S \subseteq \mathbf{R}^n$ be open, and let $f : S \rightarrow \mathbf{R}$ have continuous second partial derivatives. If $[\mathbf{x}, \mathbf{y}] \subseteq S$, then there exists $\mathbf{w} \in [\mathbf{x}, \mathbf{y}]$ such that

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{1}{2} (\mathbf{y} - \mathbf{x})^T H_f(\mathbf{w}) (\mathbf{y} - \mathbf{x}).$$

Proof of Theorem 5.3(a)

(a) $H_f(\mathbf{x})$ is positive semidefinite $\forall \mathbf{x} \in C \Leftrightarrow f$ is convex.

Proof. “ $\Rightarrow$ ”

Suppose $H_f(\mathbf{x})$ is positive semidefinite (psd) for all $\mathbf{x} \in C$. Consider $\mathbf{x}, \mathbf{y} \in C$. Then

$$\begin{aligned} f(\mathbf{y}) &= f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{1}{2} (\mathbf{y} - \mathbf{x})^T H_f(\mathbf{w}) (\mathbf{y} - \mathbf{x}), \quad \mathbf{w} \in [\mathbf{x}, \mathbf{y}] \\ &\geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \quad \text{since } H_f(\mathbf{w}) \text{ is psd.} \end{aligned}$$

By tangent plane characterization, f is convex on C .

Proof of Theorem 5.3(a) (cont'd)

(a) $H_f(\mathbf{x})$ is positive semidefinite $\forall \mathbf{x} \in C \Leftrightarrow f$ is convex.

Proof. (cont'd) “ $\Leftarrow$ ”

We proceed by contradiction. Suppose there exists $\mathbf{x}^* \in C$ such that $H_f(\mathbf{x}^*)$ is not positive semidefinite. Then the smallest eigenvalue λ of $H_f(\mathbf{x}^*)$ must be negative. Let $\mathbf{q}$ be an eigenvector corresponding to λ , normalized so that $\|\mathbf{q}\| = 1$.

Since C is open and the function $\mathbf{q}^T H_f(\mathbf{x}) \mathbf{q}$ is continuous at $\mathbf{x}^*$, there exists $r > 0$ such that $B(\mathbf{x}^*, 2r) \subseteq C$ and

$$\mathbf{q}^T H_f(\mathbf{x}) \mathbf{q} < \lambda/2 < 0, \quad \forall \mathbf{x} \in B(\mathbf{x}^*, 2r)$$

Proof of Theorem 5.3(a) (cont'd)

(a) $H_f(\mathbf{x})$ is positive semidefinite $\forall \mathbf{x} \in C \Leftrightarrow f$ is convex.

Proof. (cont'd) “ $\Leftarrow$ ”

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$$\mathbf{q}^T H_f(\mathbf{x}) \mathbf{q} < \lambda/2 < 0, \quad \forall \mathbf{x} \in B(\mathbf{x}^*, 2r)$$

Now define $\mathbf{y} := \mathbf{x}^* + r\mathbf{q}$. Then $[\mathbf{x}^*, \mathbf{y}] \subseteq B(\mathbf{x}^*, 2r) \subseteq C$, and Taylor's theorem gives

$$f(\mathbf{y}) = f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^T(r\mathbf{q}) + \frac{r^2}{2} \mathbf{q}^T H_f(\mathbf{w}) \mathbf{q} < f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^T(r\mathbf{q})$$

for some $\mathbf{w} \in [\mathbf{x}^*, \mathbf{y}]$. But the convexity of f implies that

$$f(\mathbf{y}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^T(r\mathbf{q}).$$

Thus we get a contradiction.

Take-home messages

Hessian $H_f(\mathbf{x})$	Relation	function f
positive semidefinite $\forall \mathbf{x} \in \mathcal{C}$	$\Leftrightarrow$	convex
positive definite $\forall \mathbf{x} \in \mathcal{C}$	$\Rightarrow$	strictly convex
negative semidefinite $\forall \mathbf{x} \in \mathcal{C}$	$\Leftrightarrow$	concave
negative definite $\forall \mathbf{x} \in \mathcal{C}$	$\Rightarrow$	strictly concave
indefinite at some $\hat{\mathbf{x}} \in \mathcal{C}$	$\Rightarrow$	neither convex nor concave

- Eigenvalue test.
- (Leading) principal minors.

Take-home messages (cont'd)

To verify the convexity of a function,

- $f \in C(C) \Rightarrow$ definition $(\mathbf{x}, \mathbf{y}, \lambda)$ or special operations;
- $f \in C^1(C) \Rightarrow$ tangent plane characterization $(\mathbf{x}, \mathbf{y})$;
- $f \in C^2(C) \Rightarrow$ Hessian test $(\mathbf{x})$ but we lose part of necessity!.

To verify the convexity of a set,

- definition or intersection;
- sublevel set of a convex function.

Question: how to solve a convex optimization?